

```
!pip -q install opencv-python-headless

import cv2
import numpy as np
import matplotlib.pyplot as plt

from google.colab import files

print("OpenCV version:", cv2.__version__)
```

OpenCV version: 4.14.0

```
uploaded = files.upload()

image_name = next(iter(uploaded.keys()))

image = cv2.imread(image_name)

if image is None:
    raise ValueError("The uploaded file is not a valid image.")

# Convert color image to grayscale
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

print("Image loaded successfully:", image_name)
print("Original image shape:", image.shape)
print("Grayscale image shape:", gray.shape)
```

kuchuu puchuu.jpeg
kuchuu puchuu.jpeg(image/jpeg) - 31981 bytes, last modified: 20/08/2026 - 100% done
Saving kuchuu puchuu.jpeg to kuchuu puchuu.jpeg
Image loaded successfully: kuchuu puchuu.jpeg
Original image shape: (736, 736, 3)
Grayscale image shape: (736, 736)

```
# Display the original grayscale image
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.imshow(gray, cmap="gray")
plt.title("Original Grayscale Image")
plt.axis("off")

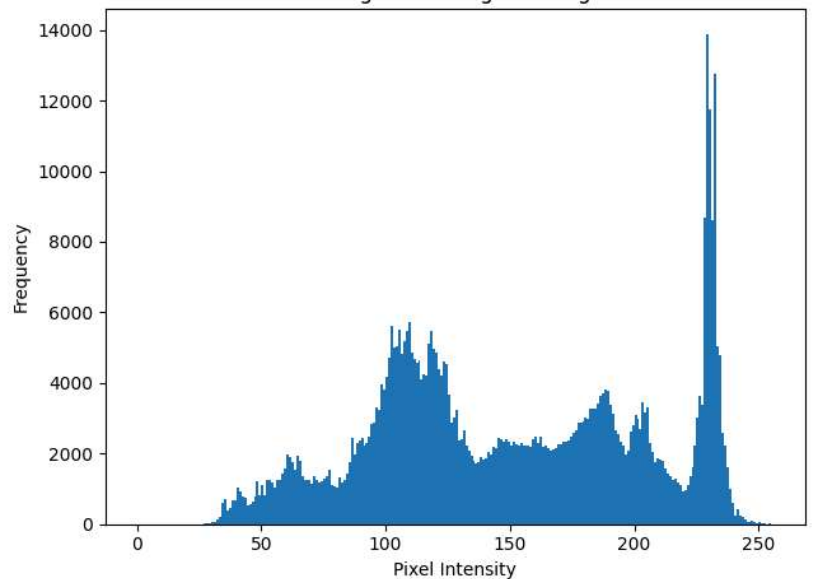
# Generate and display histogram
plt.subplot(1, 2, 2)
plt.hist(gray.ravel(), bins=256, range=[0, 256])
plt.title("Histogram of Original Image")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```

Original Grayscale Image



Histogram of Original Image



```
# Find minimum and maximum pixel intensity
min_val = np.min(gray)
max_val = np.max(gray)

# Apply contrast stretching
```

```

contrast_stretched = cv2.normalize(
    gray,
    None,
    0,
    255,
    cv2.NORM_MINMAX
)

print("Original minimum intensity:", min_val)
print("Original maximum intensity:", max_val)
print("After contrast stretching:")
print("Minimum intensity:", np.min(contrast_stretched))
print("Maximum intensity:", np.max(contrast_stretched))

# Display original and contrast-stretched images
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.imshow(gray, cmap="gray")
plt.title("Original Grayscale Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(contrast_stretched, cmap="gray")
plt.title("Contrast Stretched Image")
plt.axis("off")

plt.tight_layout()
plt.show()

```

Original minimum intensity: 25
 Original maximum intensity: 255
 After contrast stretching:
 Minimum intensity: 0
 Maximum intensity: 255

Original Grayscale Image



Contrast Stretched Image



```

# Apply Histogram Equalization
hist_equalized = cv2.equalizeHist(gray)

# Display original and histogram-equalized images
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.imshow(gray, cmap="gray")
plt.title("Original Grayscale Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(hist_equalized, cmap="gray")
plt.title("Histogram Equalized Image")
plt.axis("off")

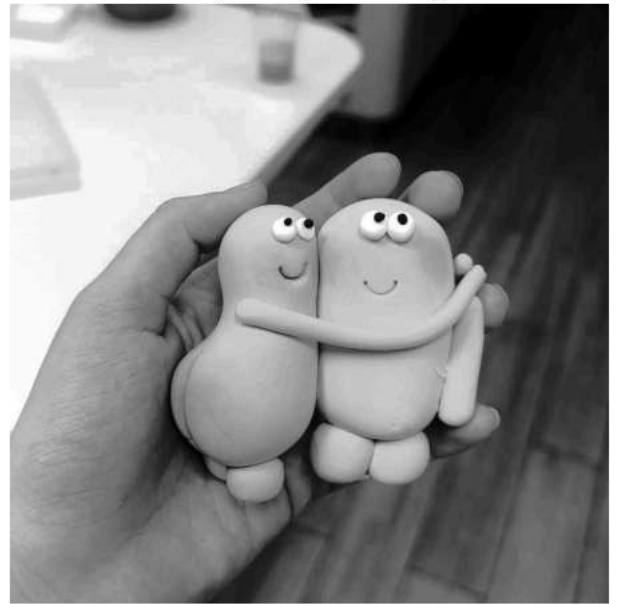
plt.tight_layout()
plt.show()

```

Original Grayscale Image



Histogram Equalized Image



```
import cv2

import numpy as np

import matplotlib.pyplot as plt

clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8,8))

clahe_img = clahe.apply(gray)

plt.figure(figsize=(5,5))

plt.imshow(clahe_img, cmap='gray')

plt.title("CLAHE Enhanced, Image")

plt.axis('off')

plt.show()
```

CLAHE Enhanced, Image



```
import cv2

import matplotlib.pyplot as plt

import numpy as np

fig, axs = plt.subplots(1, 3, figsize=(15,4))

axs[0].hist(gray.ravel(), bins=256, range=[0,256], color='gray')

axs[0].set_title("Original Histogram")

axs[1].hist(hist_equalized.ravel(), bins=256, range=[0,256], color="blue")

axs[1].set_title("Equalized Histogram")

axs[2].hist(clahe_img.ravel(), bins=256, range=[0,256], color='green')
```

```

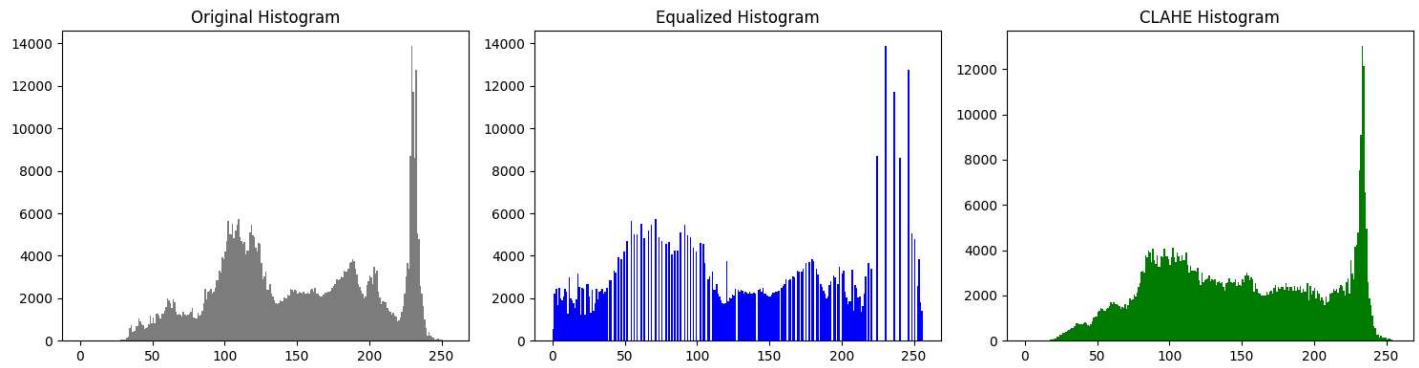
axs[2].imshow(ciahe_img, cmap=gray, bins=256, range=[0,255], color='green')

axs[2].set_title("CLAHE Histogram")

plt.tight_layout()

plt.show()

```



```

import cv2

import numpy as np

import matplotlib.pyplot as plt

fig, axs = plt.subplots(1, 4, figsize=(18,5))

axs[0].imshow(gray, cmap='gray'); axs[0].set_title("Original"); axs[0].axis('off')

axs[1].imshow(contrast_stretched, cmap='gray'); axs[1].set_title("Contrast Stretched"); axs[1].axis('off')
axs[2].imshow(hist_equalized, cmap='gray'); axs[2].set_title("Histogram Equalized"); axs[2].axis('off')
axs[3].imshow(ciahe_img, cmap='gray'); axs[3].set_title("CLAHE"); axs[3].axis('off')

plt.tight_layout()

plt.show()

```

